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$$\log_x \sqrt{8} - \log_x 4^{\frac{1}{3}} + \frac{5}{6} = 0$$

$$\Leftrightarrow \log_x \left(\frac{\sqrt{8}}{4^{\frac{1}{3}}} \right) + \frac{5}{6} = 0$$

$$\Leftrightarrow \log_x (2^{\frac{5}{6}}) + \frac{5}{6} = 0$$

$$\Leftrightarrow \log_x (2^{\frac{5}{6}}) = -\frac{5}{6}$$

$$\Leftrightarrow x^{-\frac{5}{6}} = 2^{\frac{5}{6}}$$

$$\Leftrightarrow x = 2^{\left(\frac{5}{6}\right)^{-\frac{6}{5}}}$$

$$\Leftrightarrow x = \frac{1}{2}$$

References